S2 Supplementary Data

Complete cross-carrier phase-locking tables, per-subject Rayleigh summary, both onset-hazard tests, and the full qe-interaction and SO-coupled-descriptive families

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S2.0 Complete cross-carrier pooled family (10 tests, main text R2)

Carrier	Ch.	N	R_1	R_2	T_2	perm p (R_1)	perm p (T_2)	FDR q (R_1)	FDR q (T_2)
N2-spindle onset	cardiac	23,029	0.00927	0.01508	0.00031	0.3533	0.0815	0.4417	0.1018
N2-spindle onset	resp	23,029	0.01895	0.02102	0.00080	0.1514	0.0265	0.2163	0.0378
N2-spindle peak	cardiac	23,029	0.01593	0.02005	0.00066	0.0200	0.0015	0.0333	0.0025
N2-spindle peak	resp	23,029	0.04530	0.01083	0.00217	0.0005	0.0005	0.0010	0.0010
N3-SO trough	cardiac	42,009	0.06446	0.01044	0.00426	0.0005	0.0005	0.0010	0.0010
N3-SO trough	resp	42,009	0.05347	0.01757	0.00317	0.0005	0.0005	0.0010	0.0010
N3-SO upstate	cardiac	42,009	0.06258	0.00763	0.00397	0.0005	0.0005	0.0010	0.0010
N3-SO upstate	resp	42,009	0.05036	0.00791	0.00260	0.0005	0.0005	0.0010	0.0010
REM-theta	cardiac	2,412	0.01938	0.02714	0.00111	0.4508	0.3233	0.4508	0.3423
REM-theta	resp	2,412	0.02250	0.02892	0.00134	0.4348	0.3423	0.4508	0.3423

Table 1: Complete cross-carrier pooled subject-rotation permutation family, both harmonic statistics ($n_{\text{perm}} = 2000$), BH-FDR jointly across all 10 rows within each statistic. Source: `03_multistage_phase_locking.py`, `phase_locking_pooled.csv`.

S2.1 Second-order (subject-mean) Rayleigh test, complete family

Carrier	Ch.	N subj.	Mean angle	R	Rayleigh p	FDR q
N2-spindle onset	cardiac	24	$-85.6°$	0.2729	0.1682	0.1869
N2-spindle onset	resp	24	$-131.4°$	0.4029	0.0187	0.0267
N2-spindle peak	cardiac	24	$+28.3°$	0.4622	0.00492	0.00820
N2-spindle peak	resp	24	$-127.7°$	0.6649	6.81×10^{-6}	1.70×10^{-5}
N3-SO trough	cardiac	24	$+1.6°$	0.6538	1.07×10^{-5}	2.15×10^{-5}
N3-SO trough	resp	24	$-82.8°$	0.7539	2.47×10^{-7}	2.47×10^{-6}
N3-SO upstate	cardiac	24	$-166.4°$	0.6687	5.84×10^{-6}	1.70×10^{-5}
N3-SO upstate	resp	24	$-85.3°$	0.7077	1.18×10^{-6}	5.88×10^{-6}
REM-theta	cardiac	23	$+37.4°$	0.3693	0.0417	0.0521
REM-theta	resp	23	$-178.2°$	0.2412	0.2649	0.2649

Table 2: Second-order Rayleigh test on the 24 (23 for REM-theta, one subject excluded for < 5 epochs) subject-mean directions, BH-FDR jointly across all 10 rows. Both channels' $\mu_{\text{resp,peak}}$/N2-spindle-peak rows are consistent with `S1.4`'s $\mu_{\text{resp,peak}} = -128.4°$. Source: `03_multistage_phase_locking.py`, `phase_locking_second_order.csv`.

S2.2 Per-subject Rayleigh test summary (all 5 carriers × 2 channels)

Carrier	Ch.	Median R	Median N/subj.	Range N/subj.	N subj.	N FDR-sig.
N2-spindle onset	cardiac	0.0465	1,030	161-2,562	24	0/24
N2-spindle onset	resp	0.0611	1,030	161-2,562	24	7/24
N2-spindle peak	cardiac	0.0364	1,030	161-2,562	24	1/24
N2-spindle peak	resp	0.0708	1,030	161-2,562	24	8/24
N3-SO trough	cardiac	0.0788	1,713	448-3,725	24	16/24
N3-SO trough	resp	0.0441	1,713	448-3,725	24	9/24
N3-SO upstate	cardiac	0.0875	1,713	448-3,725	24	17/24
N3-SO upstate	resp	0.0440	1,713	448-3,725	24	9/24
REM-theta	cardiac	0.1067	84	2-219	24	0/24
REM-theta	resp	0.1156	84	2-219	24	1/24

Table 3: Per-subject Rayleigh test (Zar 1999 approximation), summarized across subjects; full 237-row per-subject table (subject ID, N, mean angle, R, p, FDR q) in `phase_locking_per_subject.csv`. N2-onset/peak numbers reproduce the companion article's own S2.2 table (median R 0.046/0.061/0.036/0.071 there vs. 0.0465/0.0611/0.0364/0.0708 here - rounding only). N3-SO shows the highest per-subject FDR-survivor rate of any carrier (16-17/24), consistent with S2.0/S2.1′s ranking.

Source: `03_multistage_phase_locking.py`, `phase_locking_per_subject.csv`.

S2.3 Onset-hazard tests, full detail (main text R3)

Ch.	N cases	N ctrl.	$b_{\cos}$	$b_{\sin}$	$p_{\cos}$	$p_{\sin}$	Wald $\chi^2(2)$	Wald p
cardiac	23,029	66,827	0.0112	−0.0177	0.348	0.207	1.94	0.379
resp	23,029	66,827	−0.0217	0.0755	0.227	5.68×10^{-8}	29.48	3.96×10^{-7}

Table 4: Test A: case-control logistic regression, `is_case ~ cos(phase) + sin(phase)`, cluster-robust SE by subject (`S1.3`). BH-FDR q (2-test family): cardiac 0.379, resp 7.92×10^{-7}. Implied preferred phase: cardiac −57.8° (ns), resp +106.1°.

Source: `04_spindle_onset_hazard_model.py`, `case_control_logit.csv`.

Ch.	N	obs R_1	obs T_2	null R_1 mean	null T_2 95%	perm p (R_1)	perm p (T_2)
cardiac	23,029	0.01035	0.00034	0.00572	0.00021	0.0758	0.00399
resp	23,029	0.02145	0.00085	0.04154	0.00287	1.000	1.000

Table 5: Test B: circular time-shift surrogate ($n_{\text{perm}} = 500$, `S1.3`). BH-FDR q (2-test family, T_2): cardiac 0.00798, resp 1.000. Note the **resp** null mean R_1 (0.0415) exceeds the observed R_1 (0.0215) - the surrogate itself has higher baseline dispersion for respiration than cardiac, a property of the shifted-stream construction, not of the real data; see main text R3 for the histogram-based corroboration check that resolves which of Test A/B's disagreeing channel-specific findings is more likely genuine.

Source: `04_spindle_onset_hazard_model.py`, `circular_time_shift_surrogate.csv`.

Null construction	Cardiac p	Resp p	Construction principle
Null 0: subject-rotation permutation	0.0815	0.0265	Parametric: uniform rotation of discrete event phases
Test A: case-control logit vs. `non_event_risk_v1`	0.379	3.96×10^{-7}	Exposure-grounded: real N2 non-event time-points as controls
Test B: circular time-shift surrogate	0.0040	1.000	Exposure-grounded: shifted continuous physiological stream

Figure 1: Cross-null summary, all three onset-phase-locking tests side by side (main text Table `tab-hazard`, reproduced here with the construction-principle column). Source: `04_spindle_onset_hazard_model.py`, `cross_null_convergence.csv`.

S2.4 Complete qe-interaction family, all three operationalizations (54 tests total, main text R4)

Delta mode	Endp.	N	align. p	qe p	interaction coef	int. p	FDR q
event_control	`m`	172	0.726	0.879	$+0.081$	0.135	0.834
event_control	`d`	172	0.082	0.263	$+0.024$	0.730	0.876
event_control	`e`	172	0.326	0.845	$+0.024$	0.642	0.865
event_control	`d_n`	172	0.080	0.341	$+0.038$	0.785	0.883
event_control	`d_l`	172	0.115	0.521	$+0.008$	0.898	0.939
event_control	`d_s`	172	0.765	0.187	$+0.064$	0.440	0.865
event_control	`e_e`	172	0.632	0.303	$+0.006$	0.939	0.939
event_control	`e_s`	172	0.277	0.155	$+0.050$	0.585	0.865
event_control	`e_m`	172	0.577	0.478	$+0.082$	0.344	0.834
event_prefar	`m`	160	0.924	0.886	$+0.035$	0.651	0.865
event_prefar	`d`	160	0.023	0.199	-0.147	0.279	0.834
event_prefar	`e`	160	0.107	0.501	-0.067	0.330	0.834
event_prefar	`d_n`	160	0.096	0.256	-0.329	0.292	0.834
event_prefar	`d_l`	160	0.0030	0.389	-0.049	0.516	0.865
event_prefar	`d_s`	160	0.302	0.550	-0.112	0.371	0.834
event_prefar	`e_e`	160	0.345	0.150	$+0.044$	0.353	0.834
event_prefar	`e_s`	160	0.140	0.226	-0.221	0.249	0.834
event_prefar	`e_m`	160	0.173	0.353	-0.048	0.673	0.865

Table 6: Primary family: $\mathrm{qe} = \log(1 + \mathrm{sigma_power_z_n2})$. No interaction term survives BH-FDR (min $q = 0.834$). `align.$p$/qe $p$` are the model's main-effect p-values (not FDR-corrected, not the primary question of this family - see main text R4 for the `d`/`d_l` main-effect comparison against the companion article). Source: `05_qe_interaction_moderation.py`, `qe_interaction_primary_sigma_log.csv`.

Delta mode	Endp.	N	interaction coef	int. p	FDR q
event_control	m	172	-0.004	0.983	0.988
event_control	d	172	$+0.010$	0.952	0.988
event_control	e	172	$+0.034$	0.823	0.988
event_control	d_n	172	$+0.032$	0.878	0.988
event_control	d_l	172	$+0.019$	0.916	0.988
event_control	d_s	172	-0.366	0.134	0.988
event_control	e_e	172	$+0.203$	0.305	0.988
event_control	e_s	172	-0.085	0.679	0.988
event_control	e_m	172	-0.041	0.852	0.988
event_prefar	m	160	-0.145	0.425	0.988
event_prefar	d	160	-0.104	0.492	0.988
event_prefar	e	160	-0.031	0.707	0.988
event_prefar	d_n	160	$+0.055$	0.854	0.988
event_prefar	d_l	160	-0.220	0.307	0.988
event_prefar	d_s	160	-0.050	0.820	0.988
event_prefar	e_e	160	-0.052	0.653	0.988
event_prefar	e_s	160	$+0.003$	0.988	0.988
event_prefar	e_m	160	-0.125	0.605	0.988

Table 7: Robustness family: qe $= \log(1 + \text{amplitude})$. No interaction term survives BH-FDR (min $q = 0.988$). Source: `05_qe_interaction_moderation.py`, `qe_interaction_robust_amplitude_log.csv`.

Delta mode	Endp.	N	interaction coef	int. p	FDR q
event_control	m	172	$+0.0017$	0.252	0.725
event_control	d	172	-0.0013	0.344	0.725
event_control	e	172	-0.0009	0.384	0.725
event_control	d_n	172	-0.0017	0.507	0.725
event_control	d_l	172	-0.0011	0.434	0.725
event_control	d_s	172	-0.0001	0.942	0.942
event_control	e_e	172	-0.0003	0.793	0.840
event_control	e_s	172	-0.0012	0.487	0.725
event_control	e_m	172	$+0.0007$	0.767	0.840
event_prefar	m	160	$+0.0006$	0.695	0.834
event_prefar	d	160	-0.0025	0.243	0.725
event_prefar	e	160	-0.0007	0.589	0.757
event_prefar	d_n	160	-0.0052	0.354	0.725
event_prefar	d_l	160	-0.0013	0.276	0.725
event_prefar	d_s	160	-0.0016	0.524	0.725
event_prefar	e_e	160	$+0.0012$	0.189	0.725
event_prefar	e_s	160	-0.0034	0.312	0.725
event_prefar	e_m	160	-0.0012	0.463	0.725

Table 8: Robustness family: qe $=$ untransformed `sigma_power_z_n2`. No interaction term survives BH-FDR (min $q = 0.725$). Source: `05_qe_interaction_moderation.py`, `qe_interaction_robust_sigma_raw.csv`.

S2.5 SO-coupled descriptive-only correlations (exploratory, main text R5)

Delta mode	Endp.	N ev.	N subj.	r(delta, coupling score)	r(delta, SO amplitude)
event_control	m	21	14	0.147	−0.275
event_control	d	21	14	0.070	0.356
event_control	e	21	14	0.174	0.272
event_control	d_n	21	14	0.207	0.432
event_control	d_l	21	14	−0.165	−0.062
event_control	d_s	21	14	−0.046	0.384
event_control	e_e	21	14	0.152	−0.290
event_control	e_s	21	14	0.087	0.443
event_control	e_m	21	14	0.135	−0.038
event_prefar	m	19	13	0.114	−0.171
event_prefar	d	19	13	−0.031	0.175
event_prefar	e	19	13	0.004	0.133
event_prefar	d_n	19	13	−0.092	0.361
event_prefar	d_l	19	13	−0.017	−0.009
event_prefar	d_s	19	13	0.007	0.110
event_prefar	e_e	19	13	0.040	−0.069
event_prefar	e_s	19	13	−0.016	0.178
event_prefar	e_m	19	13	0.138	−0.106

Table 9: Descriptive Pearson correlations only (no model, no FDR family, explicitly non-confirmatory - `S1.5` part 4). Signs and magnitudes are inconsistent across endpoints (range -0.29 to $+0.44$), with no discernible pattern. N is lower than the SO-coupled subset's total (25 events/15 subjects) because each row additionally requires the relevant `delta` and covariate columns to be non-missing (the same per-mode attrition pattern as `S2.4`′s 172/160). Source: `05_qe_interaction_moderation.py`, `so_coupled_descriptive_exploratory.csv`.

S2.6 Held-out population-preferred peak phase (self-contained recomputation)

$\mu_{\text{resp,peak}}$	R (2nd-order)	N subj.	N held-out events	N excluded (geom.-valid)
−128.4° (−2.2404 rad)	0.6657	24	22,839	190

Table 10: Recomputed on this article's own `phase_resolved_v3` catalog (`S1.4`); numerically identical to the companion article's own S2.6 estimate, as expected from the bit-identity guarantee (`S1.0`) between `phase_resolved_v3` and `event_phase_v2` on all shared columns. Source: `05_qe_interaction_moderation.py` console output (deterministic given fixed input data; not separately written to CSV).

Bibliography